

Thermodynamics is the only physical theory of universal content which I am convinced, that within the framework of applicability of its basic concepts, will never be overthrown.

– Albert Einstein, Quoted in M Klein "Thermodynamics in Einstein's Thought".

1 Introduction

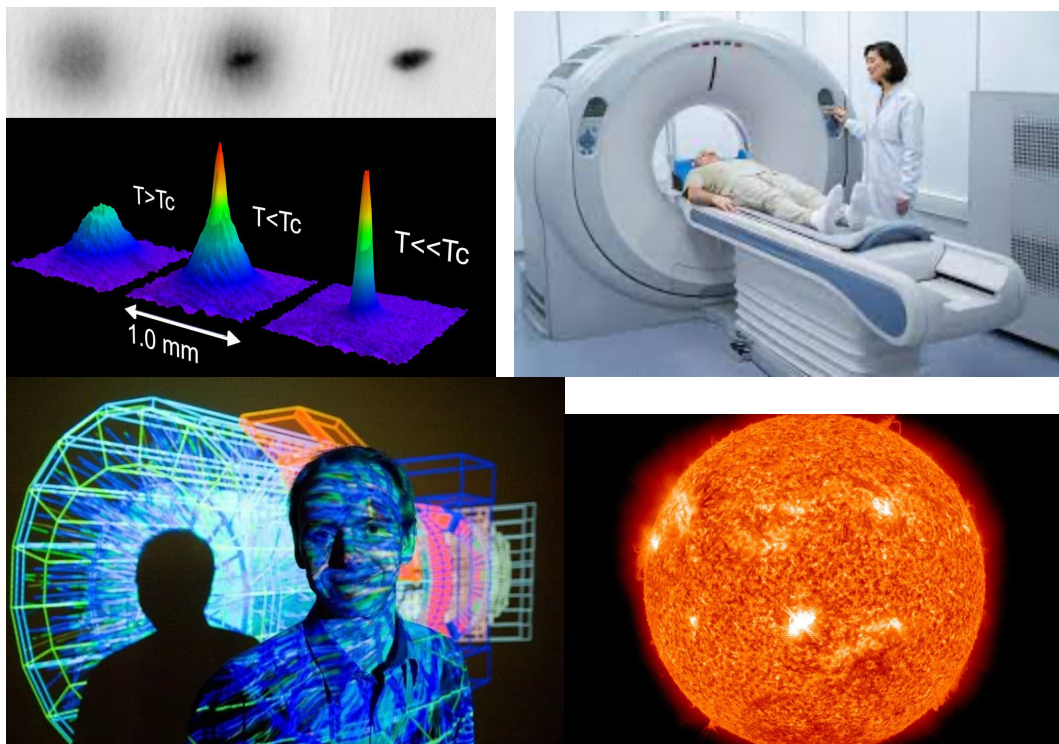


Figure 1.1: Some uses of thermal physics. Clockwise from top left: Bose-Einstein condensation of atoms at 10^{-9} K; MRI imaging; the Sun and other stars; the early Universe and the quark gluon plasma at the Alice experiment at CERN. Credits: JPL, NASA; Yashoda healthcare; National Geographic; Birmingham Mail

There are two approaches to describing macroscopic matter (Avogadro's number of particles, N_A) at a nonzero temperature.

Consider the air in a lecture theatre. We could specify the volume, the pressure and the temperature, and with greater difficulty, the number of molecules, assuming the doors and windows are shut. On the other hand we could (in principle!) specify the trajectories of the molecules, $\mathbf{r}_n(t)$, with $n = 1, 2, \dots, N_A$, or perhaps, more readily, specify their motion in a statistical manner.

These mirror the discussion in part of Y1 *Temperature and Matter*: thermodynamics and kinetic theory. We will develop both of these approaches further in this module.

Why are both needed?

Thermodynamics provides general relations between physical quantities – some easy to measure, some not – and input of data to make predictions about other measurements. This is all *without* considering (potentially unknown) microscopic details. We will see many examples of this.

On the other hand, statistical physics, or statistical mechanics, allows prediction of macroscopic measurements from a microscopic basis, providing values for quantities which can only be measured in thermodynamics. It can also provide physical insight – as in the kinetic description of pressure last year.

Entropy will be a central quantity in this module, initially as a mysteriously central entity in thermodynamics, especially in the Second Law, and then revealed as a measure of microscopic disorder in statistical physics.

1.1 Thermodynamic preliminaries

We start with defining the system we are studying, its boundary and the surroundings (the rest of the Universe) depicted in Fig. (1.2). There we see a prototypical example: the system being gas contained in a piston (constituting the boundary) with the atmosphere outside.



Figure 1.2: The system, its boundary and the surroundings. An example being gas in a piston.

The boundary is important in three ways for simple systems:

- (i) whether it allows heat to be exchanged with surroundings: *diathermal* if so, *adiabatic* if not;
- (ii) whether the boundary can move – to allow work to be done on or by the system;

- (iii) whether matter may be exchanged with surroundings, the boundary being *impermeable* if not. In this module we will almost always assume impermeability of boundaries, with the discussion of permeability being in Y3 Statistical Physics.

Thermal *equilibrium* implies at a macroscopic level the system does not vary with time. For a gas this means there are no macroscopic flows in the gas, although its constituents may be moving at a microscopic level. An implication is that the pressure in a gas is spatially uniform – otherwise flows would occur. Equilibrium is more restricted than mechanical equilibrium: most thermal systems are in stable equilibrium. *Metastable* equilibrium also occurs, with timescales from seconds to centuries. When there are no barriers to equilibration, the times to reach equilibrium can be very short: for example 10^{-12} s for electrons in a metal.

Simple thermodynamic systems usually have two variables to define their "state". For a gas they could be the pressure, p , and the volume, V , the gas occupies. For a magnetic system a natural choice is the strength of an applied field, $\mathbf{H}$, and the magnetisation, $\mathbf{M}$. For a solid one needs more parameters – the state of strain and stress. There are more complicated systems - liquid crystals and some phases of quantum gases and liquids.

1.2 Zeroth law of thermodynamics – the origin of temperature

Take two systems in equilibrium separately and bring them together, so they can exchange heat. If their states, when are in common equilibrium, are defined by (p_A, V_A) and (p_B, V_B) then we find experimentally that if one fixes, say, p_A, V_A and V_B , then one must vary p_B , to ensure equilibrium occurs (i.e. the other parameters can remain fixed). Thus there is a relation $f_{A,B}(p_A, V_A; p_B, V_B) = 0$.

Now consider three bodies, A, B and C , C can exchange energy with A and B (i.e. the walls between them are diathermal) but are together surrounded by adiabatic walls with respect to the rest of the universe.

If as a result of the diathermal walls A and C are at equilibrium with each other, and B and C are, then A and B are in equilibrium as well. This is the *zeroth law of thermodynamics* and is an experimental fact.

To bring out the consequence of this, note that if C were a thermometer, then its temperature would be that of A and B , so all three bodies are at a common temperature, T . A common temperature is an expression of bodies being in equilibrium.

What is less obvious is that this result implies that there is an *equation of state*

$$\phi(p, V) = T . \quad (1.1)$$

(See Appendix.) Although the existence of the equation of state has a thermodynamic origin, its functional form, $\phi(p, V)$, be derived within thermodynamics – it either comes from experiment, or from a calculation in statistical mechanics. Note the form of the function $\phi(p, V)$ depends on the substance – for example H_2O and Ar have very different equations of state if examined near room temperature.

In principle, the equation of state may be solved to obtain $p = p(V, T)$ which one may then plot in the p - V plane. The choice of which of p or V is naturally the "x-axis" or "y-axis" is determined by which one may vary and measure most easily. Measuring the *volume*, and controlling it (e.g. via a piston), is easier, so making V the "*independent variable*" and $p(V)$ the "*dependent variable*" is more natural.

In the Fig. (1.3) we see the *isotherms* for two different temperatures, T_1 and T_2 . To see the

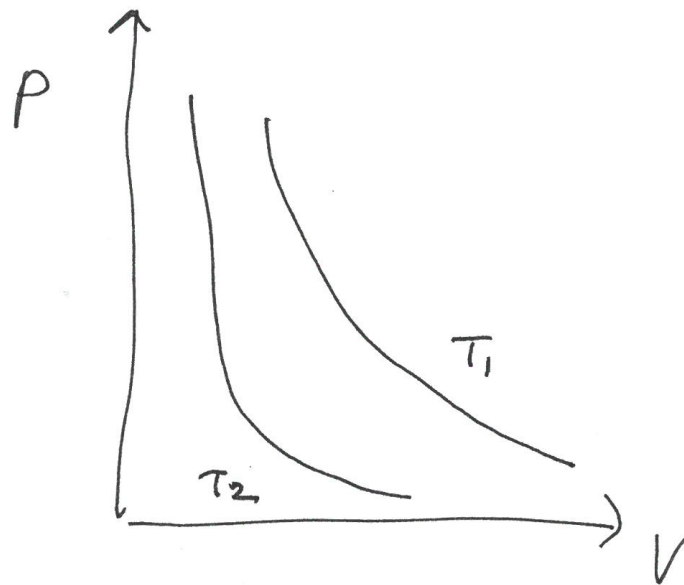


Figure 1.3: Representing the isotherms $p = p(V, T_1)$ and $p = p(V, T_2)$.

order of isotherms, note we find experimentally that for fixed V , p increases as T is increased – so the isotherms increase in T as one moves away from the V -axis.

1.3 Paths and functions of state

We have constructed isotherms in the p - V plane, which are states with the same temperature. Can we describe the system moving along them? Yes as long as we change, say, the volume in a "quasi-static" fashion. This means it is slow compared to the *relaxation time* of the system, after which the pressure and temperature are uniform across the system. In Mathematics 2B

the *heat equation* will be solved. It describes the variation of the temperature as a function of time and space, in one dimension, $T(x, t)$ in a medium obeys:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

where the constant κ is the "thermal diffusivity". By dimensional analysis, if the system has size L , then the relaxation time is

$$\tau \simeq \frac{L^2}{\kappa} ,$$

so the relaxation time grows with system size (this will be explained more clearly in Mathematics 2B). Quasi-static would mean that the fractional change in temperature over the relaxation time is small compared to the initial temperature, i.e.

$$\frac{T(t + \tau) - T(t)}{T(t)} \ll 1 .$$

Simple classes of quasi-static paths are:

- *isothermal* processes, where the system moves along an isotherm;
- *adiabatic* processes, where the system is in an adiabatic container – there is no flow of heat;
- *isobaric* processes, where the pressure is constant;
- and *isochoric* processes, where the volume is constant, so no work is performed (by a piston)¹.

A crucial important idea has slid into the discussion without its significance being observed. In the wake of the discussion of isotherms above, an unstated implication was that for any given point in the p - V plane there is a unique associated temperature, $T(p, V)$ and combining this with the discussion of paths, this would appear to be independent of path to (p, V) from, say, (p_0, V_0) .

Temperature, $T(p, V)$, is the first *function of state* we have encountered – its value being independent of path.

¹But mechanical work could be performed by other means – for example by a falling weight driving Joule's paddle wheels.

Appendices

A Equation of state from Zeroth Law (Non-examinable)

Let us consider two fluid systems, A and C , initially in equilibrium by themselves and now form a diathermal wall between them so they may exchange energy and come to a mutual equilibrium.

We cannot choose all of p_A , V_A , p_C and V_C as independent (we could choose them all independently before the diathermal contact was created). Thus there is an equation

$$F_{A,C}(p_A, V_A; p_C, V_C) = 0$$

which expresses this lack of independence. It may be solved for p_C to yield

$$p_C = f_{A,C}(p_A, V_A; V_C) . \quad (\text{A.1})$$

Now consider *three* systems, A , B and C all in equilibrium. The equilibrium between B and C provides

$$F_{B,C}(p_B, V_B; p_C, V_C) = 0 .$$

This may also be solved for p_C to yield

$$p_C = f_{B,C}(p_B, V_B; V_C) . \quad (\text{A.2})$$

We may now equate the two expressions for p_C , Eq. (A.1) and Eq. (A.2), to obtain:

$$f_{A,C}(p_A, V_A; V_C) = f_{B,C}(p_B, V_B; V_C) . \quad (\text{A.3})$$

But since we have another equation expressing that A and B are in equilibrium:

$$F_{A,B}(p_A, V_A; p_B, V_B) = 0 . \quad (\text{A.4})$$

Now Eq. (A.3) contains V_C , while Eq. (A.4) does not. So the dependence on V_C on each side of Eq. (A.3) must cancel. For example

$$f_{A,C}(p_A, V_A; V_C) = \phi_1(p_A, V_A)\zeta(V_C) + \eta(V_C)$$

and similarly for $f_{B,C}(p_B, V_B; V_C)$.

Thus we obtain

$$\phi_1(p_A, V_A) = \phi_2(p_B, V_B) .$$

and clearly by repeating this for B and C we see

$$\phi_1(p_A, V_A) = \phi_2(p_B, V_B) = \phi_3(p_C, V_C) .$$

Since the variables on each side of the above equalities are independent, each function must equal a constant, say T , the temperature. Of course it could be a function of the temperature.

Thus we arrive at the equations of state of the three systems:

$$\phi_1(p_A, V_A) = T .$$

With the obvious counterparts for B and C .